

Peyton Campbell

Software Engineer

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EXPERIENCE

OES - Full Stack Software Developer

London, ON | Aug 2025 - Present

- Own end-to-end delivery of user-facing web, mobile, desktop, and internal operations tools, translating stakeholder needs into production software for sports technology and IoT products.
- Built cross-platform .NET MAUI applications for iOS, Android, and Windows using C#, XAML, MVVM, dependency injection, Shell navigation, BLE communication, and cloud-backed data flows.
- Developed React/Next.js product configurator and admin tooling for customizable scoreboard layouts, reusable components, responsive previews, and JSON-based layout publishing workflows.
- Built Python tooling for IoT provisioning, embedded production testing, serial protocol validation, firmware programming, and cloud result logging, reducing manual factory and field workflows.
- Improved product reliability through CI/CD pipelines, automated tests, boot diagnostics, regression testing, live logging, and debugging across frontend, backend, mobile, hardware, and cloud boundaries.

PROJECTS

Scoreboard Configurator & Admin Layout Editor | *React, Next.js, TypeScript, Tailwind CSS, JSON layouts*

- Built a web-based scoreboard configurator for customer-facing product previews, including model selection, sport-specific layouts, colors, captions, team names, possession indicators, and display options.
- Refactored rendering into reusable model-specific components and editable layers for panels, LED digits, labels, logos, possession lights, arrows, and hardware indicators, improving consistency between preview and admin editing.

Provisioning Operations Console | *Python, RouterOS, LoRaWAN, TTN, PyInstaller*

- Rebuilt an internal operations console for IoT device provisioning and commissioning, guiding operators through MikroTik gateway setup, SIM activation, and Water Sniffer validation with live logs and persistent job history.
- Integrated cloud and hardware validation checks including TTN registration, join verification, boot-message detection, wet/dry test evidence polling, gateway discovery, job cancellation, and validation reporting.

ScoreController | *C#, .NET MAUI, XAML, MVVM, ReactiveUI, BLE*

- Developed a cross-platform scoreboard controller for iOS, Android, and Windows with real-time BLE communication to embedded scoreboard hardware across hockey, basketball, football, soccer, volleyball, and baseball workflows.
- Improved stability by diagnosing iOS startup crashes, refactoring MAUI resource loading, isolating page-level XAML failures, and fixing football/basketball play-clock reset edge cases.

Python Production Test Jig | *Python, serial protocols, STM32CubeProgrammer, DynamoDB, regression testing*

- Built and maintained a production test application for embedded sensor board manufacturing, integrating a custom 50-byte serial protocol, STM32 firmware programming, GUI-guided operator checks, and DynamoDB result logging.
- Fixed checksum calculation, response parsing, binary serialization, magnetic switch polarity, display validation, and firmware programming flows while adding regression tests to reduce false failures and improve traceability.

EDUCATION

University of Western Ontario - Bachelor of Computer Science

London, ON | Sep 2020 - Jan 2025

- Varsity Men's Basketball; Bob Gage Athletic Leadership Award Recipient

Fanshawe College - AI & Machine Learning Certificate

London, ON | Jan 2025 - Aug 2025

SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, HTML/CSS, Tailwind CSS, responsive UI, XAML

Backend & Data: Python, C#, .NET, REST APIs, SQL, ETL/data pipelines, DynamoDB, AWS IoT, JSON data flows

Infra & Tools: AWS, Azure DevOps, CI/CD, Docker, Git, PyInstaller, unit testing, regression testing, Agile

Systems: Bluetooth Low Energy, serial protocols, LoRaWAN, MikroTik RouterOS, STM32 firmware programming